

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title: Toy World – E-commerce Database Management System

1. Database Design

1.1 Category Table

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

1.2 Product Table

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Category_ID	Not Null
Price	Not Null
Stock	-
Brand	-
Description	-

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: One-to-Many (1:M) — one category contains many products.

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(50) UNIQUE,  
4     Description VARCHAR2(200)  
5 );  
  
Table created.
```

The Category table is created with Category_ID as the Primary Key and Category_Name as a Unique Key.

3.2 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID NUMBER PRIMARY KEY,  
3     Product_Name VARCHAR2(100) NOT NULL,  
4     Category_ID NUMBER NOT NULL,  
5     Price NUMBER(10,2) NOT NULL,  
6     Stock NUMBER,  
7     Brand VARCHAR2(50),  
8     Description VARCHAR2(200),  
9     CONSTRAINT fk_product_category  
10    FOREIGN KEY (Category_ID)  
11    REFERENCES Category(Category_ID)  
12 );  
  
Table created.
```

The Product table is created with Product_ID as the Primary Key and Category_ID as a Foreign Key referencing the Category table.

4. Data Insertion

4.1 Category Records

Ten category records are inserted into the Category table.

```
SQL> INSERT INTO Category VALUES (1, 'Educational Toys', 'Toys that support learning and education');
1 row created.

SQL> INSERT INTO Category VALUES (2, 'Soft Toys', 'Soft and cuddly toys for children');
1 row created.

SQL> INSERT INTO Category VALUES (3, 'Remote Control Toys', 'Toys operated using remote control');
1 row created.

SQL> INSERT INTO Category VALUES (4, 'Board Games', 'Games played using boards and pieces');
1 row created.

SQL> INSERT INTO Category VALUES (5, 'Outdoor Toys', 'Toys suitable for outdoor activities');
1 row created.

SQL> INSERT INTO Category VALUES (6, 'Dolls', 'Dolls and doll accessories');
1 row created.

SQL> INSERT INTO Category VALUES (7, 'Building Blocks', 'Blocks used for building and construction');
1 row created.

SQL> INSERT INTO Category VALUES (8, 'Puzzles', 'Puzzles for logical thinking and problem solving');
1 row created.

SQL> INSERT INTO Category VALUES (9, 'Action Figures', 'Character-based action figures');
1 row created.

SQL> INSERT INTO Category VALUES (10, 'Musical Toys', 'Toys that produce musical sounds');
1 row created.
```

4.2 Product Records

Ten product records are inserted into the Product table. Each product is linked to a category using Category_ID.

```

SQL> INSERT INTO Product VALUES (101, 'ABC Learning Kit', 1, 450, 25, 'FunLearn', 'Educational learning kit');
1 row created.

SQL> INSERT INTO Product VALUES (102, 'Teddy Bear', 2, 550, 20, 'ToyLand', 'Soft teddy bear');
1 row created.

SQL> INSERT INTO Product VALUES (103, 'Remote Car', 3, 1200, 15, 'SpeedX', 'Remote controlled car');
1 row created.

SQL> INSERT INTO Product VALUES (104, 'Chess Board', 4, 650, 18, 'Classic Games', 'Wooden chess board');
1 row created.

SQL> INSERT INTO Product VALUES (105, 'Football', 5, 400, 30, 'PlayPro', 'Outdoor football');
1 row created.

SQL> INSERT INTO Product VALUES (106, 'Barbie Doll', 6, 850, 12, 'DollWorld', 'Fashion doll');
1 row created.

SQL> INSERT INTO Product VALUES (107, 'Building Blocks Set', 7, 700, 22, 'BuildMax', 'Colorful building blocks');
1 row created.

SQL> INSERT INTO Product VALUES (108, 'Jigsaw Puzzle', 8, 350, 28, 'PuzzlePro', 'Kids jigsaw puzzle');
1 row created.

SQL> INSERT INTO Product VALUES (109, 'Superhero Figure', 9, 600, 16, 'HeroToys', 'Action figure');
1 row created.

SQL> INSERT INTO Product VALUES (110, 'Mini Piano', 10, 900, 10, 'MusicKids', 'Toy musical piano');
1 row created.

```

5. Data Retrieval

5.1 Display Category Records

```

SQL> SELECT*FROM Category;

```

CATEGORY_ID	CATEGORY_NAME	DESCRIPTION
1	Educational Toys	Toys that support learning and education
2	Soft Toys	Soft and cuddly toys for children
3	Remote Control Toys	Toys operated using remote control
4	Board Games	Games played using boards and pieces
5	Outdoor Toys	Toys suitable for outdoor activities
6	Dolls	Dolls and doll accessories
7	Building Blocks	Blocks used for building and construction
8	Puzzles	Puzzles for logical thinking and problem solving
9	Action Figures	Character-based action figures
10	Musical Toys	Toys that produce musical sounds

```

10 rows selected.

```

All category records are retrieved to verify successful insertion.

5.2 Display Product Records

```
SQL> SELECT * FROM Product;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	PRICE	STOCK	BRAND	DESCRIPTION
101	ABC Learning Kit	1	450	25	FunLearn	Educational learning kit
102	Teddy Bear	2	550	20	ToyLand	Soft teddy bear
103	Remote Car	3	1200	15	SpeedX	Remote controlled car
104	Chess Board	4	650	18	Classic Games	Wooden chess board
105	Football	5	400	30	PlayPro	Outdoor football
106	Barbie Doll	6	850	12	DollWorld	Fashion doll
107	Building Blocks Set	7	700	22	BuildMax	Colorful building blocks
108	Jigsaw Puzzle	8	350	28	PuzzlePro	Kids jigsaw puzzle
109	Superhero Figure	9	600	16	HeroToys	Action figure
110	Mini Piano	10	900	10	MusicKids	Toy musical piano

```
10 rows selected.
```

All product records are retrieved to verify successful insertion.

6. Product Update

The price of a product is updated using its Product_ID.

```
SQL> UPDATE Product
2 SET Price = 1350
3 WHERE Product_ID = 103;

1 row updated.

SQL> UPDATE Product
2 SET Price = 750
3 WHERE Product_ID = 104;

1 row updated.

SQL> UPDATE Product
2 SET Price = 450
3 WHERE Product_ID = 105;

1 row updated.
```

```
SQL> SELECT*FROM Product;
```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	PRICE	STOCK	BRAND	DESCRIPTION
101	ABC Learning Kit	1	450	25	FunLearn	Educational learning kit
102	Teddy Bear	2	550	20	ToyLand	Soft teddy bear
103	Remote Car	3	1350	15	SpeedX	Remote controlled car
104	Chess Board	4	750	18	Classic Games	Wooden chess board
105	Football	5	450	30	PlayPro	Outdoor football
106	Barbie Doll	6	850	12	DollWorld	Fashion doll
107	Building Blocks Set	7	700	22	BuildMax	Colorful building blocks
108	Jigsaw Puzzle	8	350	28	PuzzlePro	Kids jigsaw puzzle
109	Superhero Figure	9	600	16	HeroToys	Action figure
110	Mini Piano	10	900	10	MusicKids	Toy musical piano

```
10 rows selected.
```

7. Product Deletion

Selected products are deleted from the Product table using their Product_ID.

```

SQL> DELETE FROM Product
      2 WHERE Product_ID = 106;

1 row deleted.

SQL> DELETE FROM Product
      2 WHERE Product_ID = 107;

1 row deleted.

SQL> DELETE FROM Product
      2 WHERE Product_ID = 110;

1 row deleted.

```

```

SQL> SELECT*FROM Product;

```

PRODUCT_ID	PRODUCT_NAME	CATEGORY_ID	PRICE	STOCK	BRAND	DESCRIPTION
101	ABC Learning Kit	1	450	25	FunLearn	Educational learning kit
103	Remote Car	3	1350	15	SpeedX	Remote controlled car
104	Chess Board	4	750	18	Classic Games	Wooden chess board
105	Football	5	450	30	PlayPro	Outdoor football
108	Jigsaw Puzzle	8	350	28	PuzzlePro	Kids jigsaw puzzle
109	Superhero Figure	9	600	16	HeroToys	Action figure

```

6 rows selected.

```

8. Category-wise Product Reports

8.1 Product Listing by Category

This report displays the Category Name along with the Product Name, Price, Stock, and Brand belonging to each category. An INNER JOIN is used between the Category and Product tables.

```

SQL> SELECT
      2 C.Category_ID,
      3 C.Category_Name,
      4 P.Product_ID,
      5 P.Product_Name,
      6 P.Price,
      7 P.Stock,
      8 P.Brand
      9 FROM Category C
     10 INNER JOIN Product P
     11 ON C.Category_ID = P.Category_ID
     12 ORDER BY C.Category_ID;

```

CATEGORY_ID	CATEGORY_NAME	PRODUCT_ID	PRODUCT_NAME	PRICE	STOCK	BRAND
1	Educational Toys	101	ABC Learning Kit	450	25	FunLearn
3	Remote Control Toys	103	Remote Car	1350	15	SpeedX
4	Board Games	104	Chess Board	750	18	Classic Games
5	Outdoor Toys	105	Football	450	30	PlayPro
8	Puzzles	108	Jigsaw Puzzle	350	28	PuzzlePro
9	Action Figures	109	Superhero Figure	600	16	HeroToys

```

6 rows selected.

```

This query displays the products under their respective categories and demonstrates the one-to-many relationship between Category and Product.

Conclusion

The Category and Product tables were successfully designed and created. Primary and foreign key relationships were established, sample records were inserted, and CRUD operations such as Insert, Select, Update, and Delete were performed. An INNER JOIN was used to generate a category-wise product report, demonstrating the one-to-many relationship between Category and Product.